



Comparison of Hydrogen Production Models for Hamburg Airport

Hydrogen and Sustainable Aviation Fuels (SAF) – Key Factors for Carbon-Free Aviation

GROUP 2: PRODUCTION

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1 Wind Energy Potential as a key energy source for local hydrogen production for Hamburg Airport

1.1 Introduction

To meet the challenge of producing and ensuring a continuous supply of such a large volume of hydrogen for Hamburg Airport (projected at 6,045 GWh/year), the supply strategy must be diversified. Accordingly, this analysis focuses on the potential of onshore wind power, which is currently the most widely-used source of green energy in northern Germany. This approach is in line with the strategy of the airport itself, which is already developing its own Heidmoor wind farm project with a planned capacity of 36 MW. This scenario, therefore, assesses not only the potential of such in-house projects but, more importantly, explores the possibilities of the vast wind energy resources available in Hamburg's immediate vicinity as the key to meeting full demand in the future.

1.2 Analysis of the current situation

1.2.1 Available Assets

The first key element is a geographic analysis of the distribution of wind turbines, shown in the map below (Fig. 1).

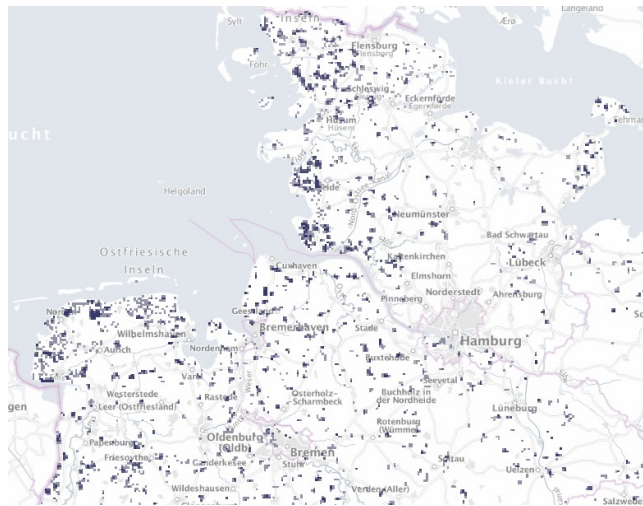


Figure 1: Wind turbine density distribution in Germany (Retrieved from Geoportal.de, [1])

Table 1: Cumulative capacity and number of wind turbines in the German federal states (2024-12-31)[5]

Federal State	Cumulative Capacity (MW)	Cumulative Number	Share	Capacity per Area (kW/km ²)
Lower Saxony	12,967	6,123	20%	272
Brandenburg	8,992	4,079	14%	284
Schleswig-Holstein	8,984	3,266	14%	568
North Rhine-Westphalia	7,800	3,222	12%	229
Saxony-Anhalt	5,512	2,199	9%	264
Rhineland-Palatinate	4,198	1,781	7%	210
Mecklenburg-Western Pomerania	3,795	1,844	6%	120
Bavaria	2,679	1,152	4%	37
Hesse	1,895	800	3%	89
Baden-Württemberg	1,856	800	3%	52
Thuringia	1,835	867	3%	114
Saxony	1,220	553	2%	66
Saarland	553	209	1%	211
Bremen	427	116	1%	683
Hamburg	125	67	0%	166
Berlin	26	7	0%	28
Germany	63,551	28,717	100%	178

The map clearly illustrates that Hamburg's territory is virtually devoid of onshore wind turbines. At the same time, the surrounding regions-Schleswig-Holstein in the north and Lower Saxony in the south-have the highest concentration of wind power in all of Germany. This geographic proximity makes importing energy from these regions the most logical and efficient solution. The conclusions of the map analysis are directly confirmed by the figures. Table 1. shows the cumulative installed capacity in each state (as of Dec. 31, 2024).

The data shows that the regions neighbouring Hamburg are the absolute leaders in Germany:

- Schleswig-Holstein: 8,992 MW of installed capacity.
- Lower Saxony: 12,001 MW of installed capacity.

The combined capacity of these two regions, in excess of 21 GW, exceeds the airport's demand many times over and represents a huge available reservoir of green energy. The high power density ratio (368 kW/km² and 251 kW/km², respectively) further demonstrates the maturity of these markets and the availability of infrastructure.

1.2.2 Untapped Potential

In addition to the installed capacity itself, a key phenomenon is the growing volume of curtailed energy. The chart at Figure 2. illustrates the amount of wind energy that is produced but intentionally not fed into the grid due to overloading or lack of demand.

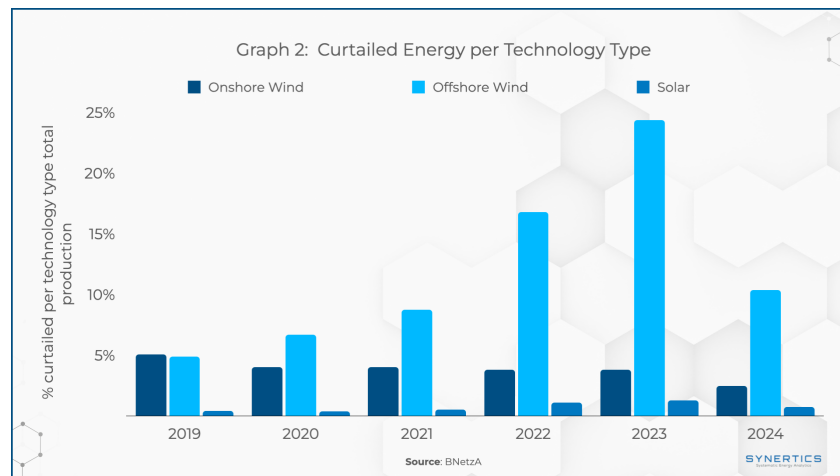


Figure 2: Curtailed Energy per Technology Type [2].

The data specifies that production from onshore wind farms in Germany has been reduced by 3384 GWh in 2024 alone. Although this is a 15% decrease from 2023, the volume still represents a huge untapped energy resource. By comparison, it is an amount of energy equivalent to more than half of Hamburg Airport's annual hydrogen demand projected for 2040+ (6,045 GWh) and nearly 100% for 2035.

1.3 Required Energy Capacity

To produce a certain amount of energy (in GWh), a certain amount of installed capacity (in MW) is needed. The table below shows how much capacity is needed in selected technologies to meet the airport's projected energy needs in key years. To accurately estimate how much power is needed to produce the required amount of energy, the scenario makes a number of technological assumptions for each renewable source.

For onshore wind power, it was assumed that the turbines would operate in conditions typical of areas near Hamburg, where the average wind speed at a height of 150 metres is 9 m/s. The model was based on the installation of turbines of 3.3 MW each, for which an average annual capacity factor of 30% was assumed. More favourable conditions were assumed for offshore wind power, which complements the mix. The model assumes a higher average wind speed, reaching 10.5 m/s, and the use of much larger 15 MW turbines. This translates into a higher and more stable capacity factor, which was estimated at 45%. In addition, to make the model more realistic, the current percentage of

energy curtailment for each technology was included. It was assumed that curtailment losses would be 3% for onshore farms, 10% for offshore farms and 1% for photovoltaic installations.

The table 2 shows the results of these calculations, i.e. the required installed capacity for each technology for each time horizon.

Table 2: Energy projections for Northern Germany, Source: Wind data: Global Wind Atlas[6], Solar data: PVGIS Tool [7], Power coefficients: [8]

Year	Energy Demand (GWh)	Onshore Wind (MW)	Offshore Wind (MW)	Solar Energy (MWp)
2025	390	153	110	394
2030	2080	816	586	2101
2035	3510	1377	989	3545
2040	4550	1785	1282	4596
2040+	6045	2371	1704	6105

The results do not represent an energy mix but are a comparison of three independent technology paths. They make it possible to assess the scale of investment and infrastructure needs for each option separately.

1.4 Energy prices and forecasting

This section presents a detailed annual financial forecast for a hydrogen production scenario from 2025 to 2040. This analysis provides an understanding of how key factors – hydrogen demand, energy demand and price – affect the final energy cost of producing one kilogram of hydrogen. The table below shows a detailed annual financial forecast for the hydrogen production scenario, taking into account projected demand and energy prices.

Year	2025	2026	2027	2028	2029	2030	2031	2032
H2 Demand (t)	6000	11200	16400	21600	26800	32000	36400	40800
Energy Demand (MWh)	92308	172308	252308	332308	412308	492308	560000	627692
Energy Price (EUR)	101	107	98	95	85	80	83	82
Energy costs (kEUR)	9366	18394	24729	31519	35232	39276	46301	51251
EUR per kg H2	1.56	1.64	1.51	1.46	1.31	1.23	1.27	1.26

Year	2033	2034	2035	2036	2037	2038	2039	2040
H2 Demand (t)	45200	49600	54000	57200	60400	63600	66800	70000
Energy Demand (MWh)	695385	763077	830769	880000	929231	978462	1027692	1076923
Energy Price (EUR)	77	78	75	73	75	80	80	83
Energy costs (kEUR)	53649	59535	62665	64451	69729	78023	82483	89083
EUR per kg H2	1.19	1.20	1.16	1.13	1.15	1.23	1.23	1.27

Table 3: Energy capacity projections for Northern Germany (Distributed Energy scenario) 2025-2040, [9]

The forecast shows that despite the huge increase in production, the unit energy cost of producing hydrogen is on a downward trend over the next decade. However, it should be emphasised that the values shown (1.15-1.56€/kg) reflect only the cost of purchasing electricity. The final price of green hydrogen must additionally take into account the depreciation (CAPEX) and operating costs (OPEX) of the electrolyser, water treatment, storage and distribution costs, as will be presented in the next section.

1.5 Levelized Cost of Hydrogen

To get the full economic picture and be able to compare different scenarios, it is necessary to calculate the Levelized Cost of Hydrogen (LCOH). This is a standard, internationally recognised methodology that takes into account all costs incurred throughout the project life cycle:

- Capital expenditures (CAPEX)
- Operating costs (OPEX)
- Cost of electricity.
- Cost of capital (investment financing).

To illustrate this methodology and the main cost components, the following Figure 3 is a sample calculation made with the LCOH Calculation Tool, developed by the renowned organisations Agora Energiewende and umlaut (part of Accenture). This dashboard allows for an in-depth analysis of the cost structure and its sensitivity to changes in key input parameters.

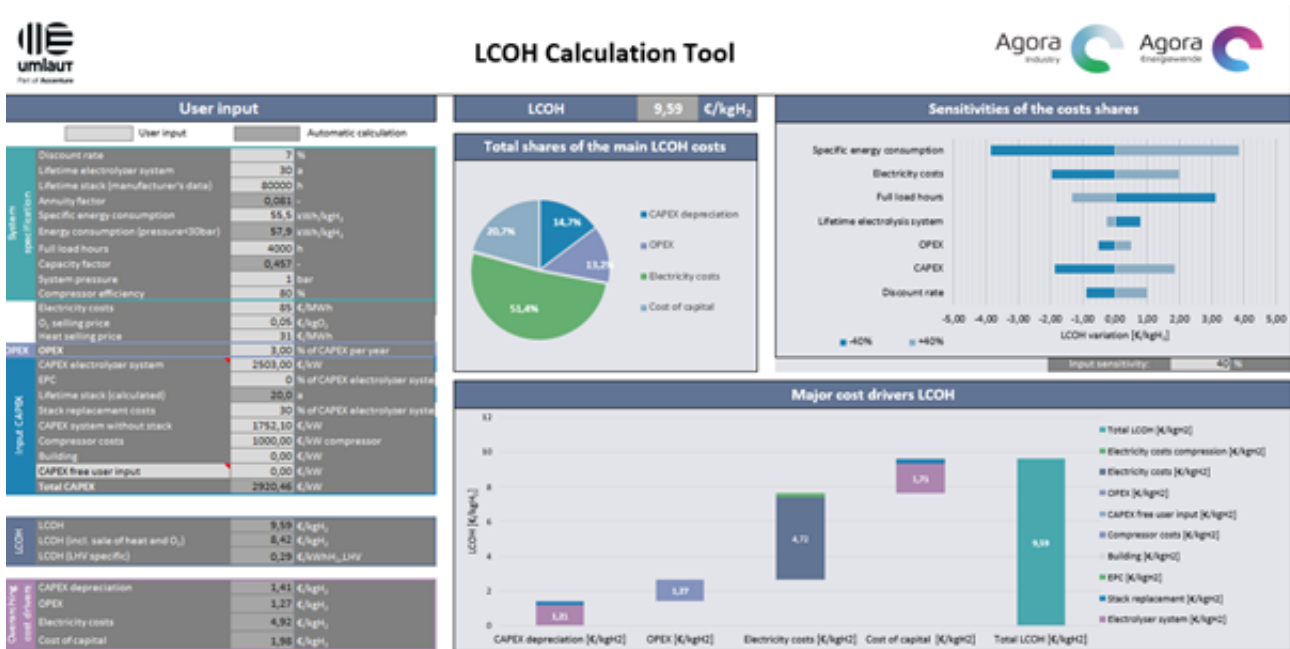


Figure 3: LCOH dashboard for local scenario, source: [3]

Key Findings from the Analysis:

- Energy is more than half the total cost: the cost of electricity accounts for 51.4% of the final price of hydrogen. Providing low-cost energy is absolutely key.
- Investment is 1/3 of the total cost: The costs associated with the investment (CAPEX) and its financing (cost of capital) total more than 35% of the price. This is the second most important factor after energy.
- Operating time is critical: The price of hydrogen is most sensitive to the price of electricity and the annual number of operating hours. The longer an electrolyser runs, the cheaper the hydrogen is.
- The price of the electrolyser is less important: The purchase cost of the electrolyser itself (CAPEX) has less impact on the final price than the price of energy and operating hours.
- The baseline result is 9.59€/kg

1.6 Hydrogen production and transport facilities

The success of the hydrogen strategy for Hamburg Airport is inextricably linked to the development of regional infrastructure for hydrogen production and transportation. Hamburg, as one of the

future leaders of Europe's hydrogen economy, is investing heavily in building a complete value chain, which creates favourable conditions for such large-scale customers as the airport. The development of hydrogen infrastructure in northern Germany is best illustrated by the map on Figure 4, which shows the density of planned and ongoing projects.

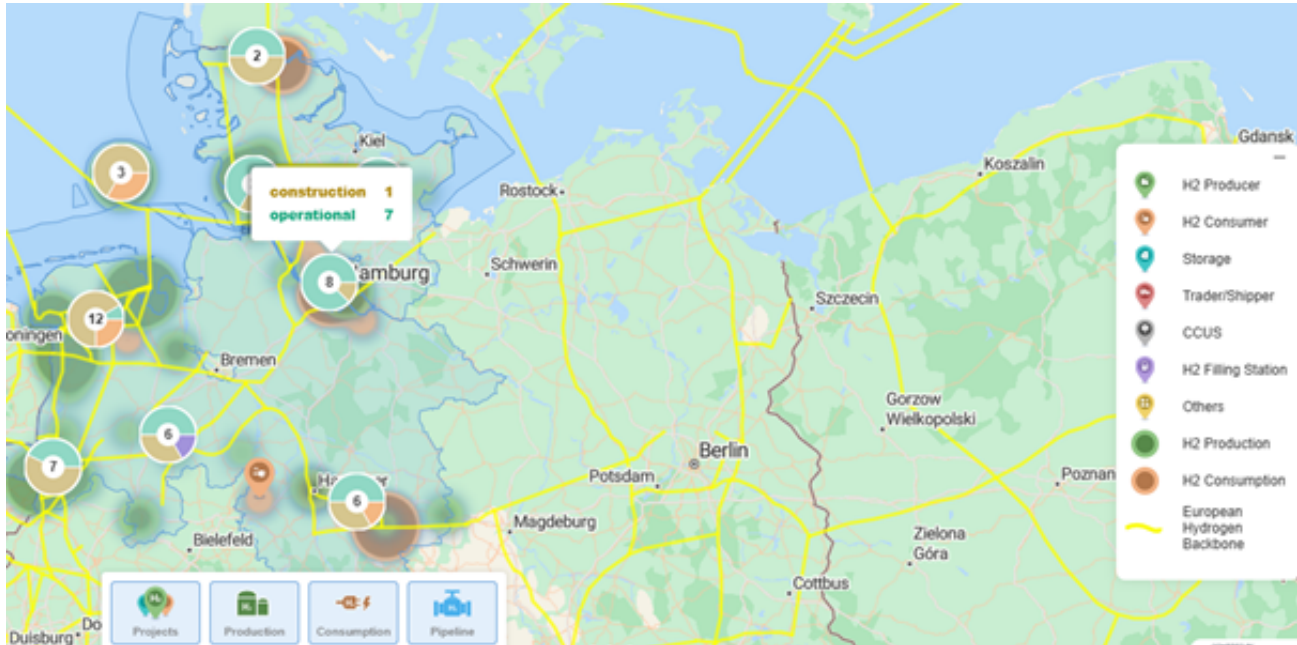


Figure 4: Hydrogen supply chain in Northern Germany, source: [4]

An analysis of the hydrogen infrastructure map clearly confirms that the Hamburg region is a centre of intensive development for the entire hydrogen supply chain. Key ongoing projects in the immediate vicinity include HH-WIN in Hamburg (100 MW), Westküste100 in Heide, where 30 MW is already in operation with plans to expand to 700 MW, and the SALCOS steel mill decarbonisation project in Salzgitter (100 MW+). The total capacity of the planned plants in the region is counted in gigawatts, demonstrating the enormous scale of the investments. The visualisation is complemented by the course of the European Hydrogen Backbone, marked on the map with yellow lines. It illustrates how Hamburg's local distribution network will be integrated into the national and European transmission infrastructure, enabling large-scale hydrogen transportation. The commissioning of this strategic link is planned for 2027-2028, which means that during the interim period, before the pipeline infrastructure is commissioned, the only available option for supplying the airport with hydrogen will be by road.

Due to limitations in rooftop-level detail (e.g., slope and usable surface area), this proposal is framed as a forward-looking recommendation, grounded in spatial analysis but requiring further refinement through technical and financial studies.

2 Rooftop Solar Potential as an Integrated Future Energy Source for Hamburg Airport

2.1 Introduction

Hamburg Airport is projected to require up to **6,045 GWh/year** of hydrogen energy by 2040, as identified in our calculations of projected hydrogen demand. While offshore wind remains a cornerstone of the hydrogen supply strategy, this study explores the complementary role of **rooftop solar photovoltaics (PV)** — particularly across residential rooftops and airport infrastructure — as a decentralised and integrated solution to support hydrogen production.

Due to limitations in rooftop-level detail (e.g., slope and usable surface area), this proposal is framed as a forward-looking recommendation, grounded in spatial analysis but requiring further refinement through technical and financial studies.

2.2 Geospatial Methodology: Rooftop Screening via QGIS

We conducted a **GIS-based screening analysis** using open geospatial data to identify rooftops suitable for solar panel deployment near Hamburg Airport. While the analysis was constrained by the absence of detailed roof geometry, the following method was used to perform a reliable first-stage feasibility assessment.

2.2.1 Data Inputs

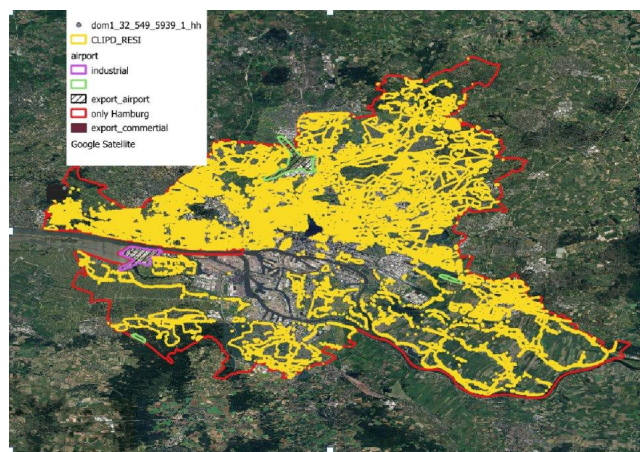


Figure 5: Map of Residential and Airport Zones in Hamburg Used for Rooftop Solar Screening. Yellow areas represent residential buildings; the striped area shows the airport boundary. Industrial and commercial zones were excluded. This classification supported rooftop selection for solar suitability analysis.

Table 4: Dataset descriptions and specifications

Dataset	Description	Formatting	Purpose
Building footprints	Polygon shapefile of rooftops (residential and airport)	Vector	Building footprints are saved as vector polygons because they show each roof's exact shape and size. This makes it easy to measure roof area, direction, and tilt. These footprints are the main data used to choose which roofs are good for solar panels.
Digital Surface Model (DSM)	Elevation model including roof structures	Raster (GeoTIFF)	The DSM provides 3D elevation data needed to derive roof slope and aspect. Raster is used because elevation is continuous over space and is best stored as gridded data for surface analysis and slope/aspect extraction in GIS.
Satellite imagery	Bing or Google imagery for visual validation	Raster overlay	High-resolution images are used as a background map to check that the building footprints match real roofs. They help make sure the shapes are correct and show what is around the roofs. Photos are saved as raster files and can be easily added under the vector layers for checking.
(Optional) Solar radiation layer	Modelled insolation for prioritization	Raster	Solar radiation shows how much sunlight different parts of a roof get. It is saved as a raster so it can show this information for every small area and be combined with the roof shapes to find the best roofs for solar panels.

The map in Fig. 5 illustrates the spatial scope used for rooftop solar analysis. Residential zones were identified from land use data and clipped to Hamburg’s boundary, while the airport area was manually outlined. Industrial and commercial areas were excluded to focus the study on residential and airport rooftops only, aligning with the intended alternative energy recommendation.

2.2.2 Recommended Future Selection Criteria for Rooftop Solar Assessment

While this study successfully identified residential and airport rooftop zones, detailed roof geometry data (e.g., roof angle and usable surface area) was not available at sufficient resolution to complete rooftop-level analysis. For future assessments, we recommend applying practical geospatial selection criteria consistent with established rooftop solar potential studies ([10]). Specifically, rooftops with moderate slopes suitable for effective PV mounting, aspects between 135° and 225° to target south-facing surfaces in the Northern Hemisphere, and a minimum usable area large enough to accommodate a typical residential PV system should be prioritised. Additionally, areas with higher solar radiation potential should be considered to maximise energy yield. These practical parameters provide a technically sound basis for detailed feasibility screening once higher-resolution elevation data becomes available.

Two external examples reinforce this strategy. First, Airsight GmbH [11] demonstrates the feasibility of rooftop and ground-mounted PV at airports, highlighting how careful siting, sufficient usable area, and compliance with technical constraints (such as glare and safety) can deliver significant capacity — with estimates up to 5–15 MWp/km², supporting this study’s airport solar potential estimates. Second, Hamburg’s Senate (Gramwzielone.pl, [11]) has introduced fast-track policies to expand rooftop PV by targeting technically suitable roofs on homes and public buildings. These policies aim for over 300 MWp by 2025 and 500 MWp by 2030, directly aligning with the residential rooftop zones and minimum size thresholds recommended here. Together, these cases confirm that clearly defined selection parameters and site-specific technical feasibility are essential to accelerate solar deployment.

2.3 Conclusion

This initial rooftop solar assessment shows that residential and airport rooftops around Hamburg Airport have strong potential to complement the city’s hydrogen and renewable energy goals. Although limited by available roof geometry data, the GIS-based screening provides a reliable first step to identify promising zones and guide further detailed analysis. By applying clear selection criteria and using practical geospatial datasets, this approach sets the groundwork for technical studies, investment planning, and policy action to unlock rooftop PV as an integrated part of Hamburg’s future energy mix.

3 Import Scenario

3.1 Hydrogen Production in Morocco

3.1.1 Green Hydrogen (solar/wind electrolysis):

Morocco benefits from excellent renewable resources, with solar irradiation reaching approximately 2,200 kWh/m²/year and average wind speeds ranging from 7.5 to 10 m/s [12]. Hydrogen production via electrolysis typically requires 50–60 kWh of electricity per kilogram of H₂, depending on the technology used [13]. Current production costs in Morocco range from €1.8 to €2.6 per kg in optimal zones [14], and between €2.8 and €5.6 per kg on average [15].

These costs are strongly influenced by the capacity factor, which helps reduce electricity cost. However, due to renewable intermittency (day/night cycle and seasonal variation), energy storage systems are often required. By 2030, projected production costs are expected to be around €3.1 to €3.2 per kg, with a potential decrease below €2/kg by 2050 as technologies mature and scale [16].

3.1.2 Blue Hydrogen (natural gas + CO₂ capture):

Produced via steam methane reforming (SMR) with carbon capture (CCS), blue hydrogen requires approximately 49 kWh of natural gas plus 0.8 kWh of electricity per kilogram of H₂ [17]. Costs typically range from €2.6 to €3.3/kg, depending on gas prices and CCS investment [15].

Since Morocco imports its natural gas, costs are slightly higher, but blue hydrogen remains cheaper than green hydrogen for now. Climate conditions have little impact on the process, and future cost reductions will depend on improved carbon capture technologies.

3.1.3 Grey Hydrogen (natural gas, no capture):

Conventional SMR without CCS consumes approximately 44.5 kWh of natural gas per kilogram of H₂ [17]. It's the cheapest option today (€0.5–1.6/kg), but also the most carbon-intensive [15].

In Morocco, it would rely on imported gas, which may increase vulnerability to global price volatility. Due to its environmental impact, grey hydrogen is expected to be phased out by 2030–2050, especially as carbon pricing increases [13].

The chart (Fig 6.) illustrates hydrogen production costs in Morocco, comparing three production pathways: green, blue, and grey hydrogen. It presents both current costs and projections for 2050. As of today, green hydrogen produced via electrolysis using renewable energy costs around €2.6 per kilogram [17]. However, this cost is expected to drop to €2.0 per kilogram by 2050, due to anticipated reductions in electrolyzer costs, increased efficiency, and Morocco's strong solar and wind potential [13].

Blue hydrogen, derived from natural gas with carbon capture, is currently the most expensive option at €3.3/kg. Even by 2050, its cost is projected to remain relatively high (€2.5/kg) because of the added expenses related to carbon capture and storage (CCS). Grey hydrogen, produced from fossil

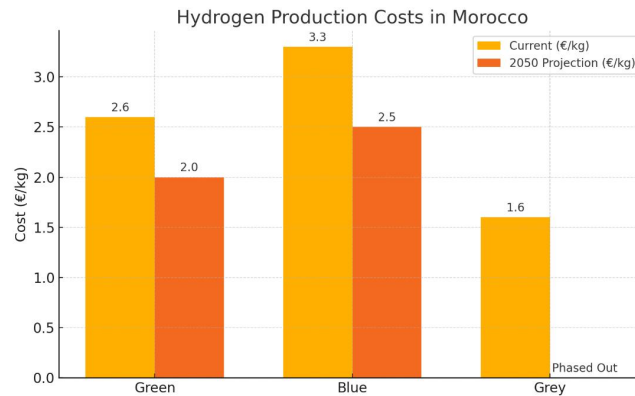


Figure 6: Hydrogen Production Costs in Morocco (Current vs 2050 Projection)

fuels without CCS, remains the cheapest today at €1.6/kg, but it is set to be phased out due to its high CO₂ emissions and incompatibility with decarbonization goals.

This cost trend highlights Morocco's strategic potential as a competitive hub for green hydrogen production, especially as global markets shift towards sustainable and low-carbon energy sources.

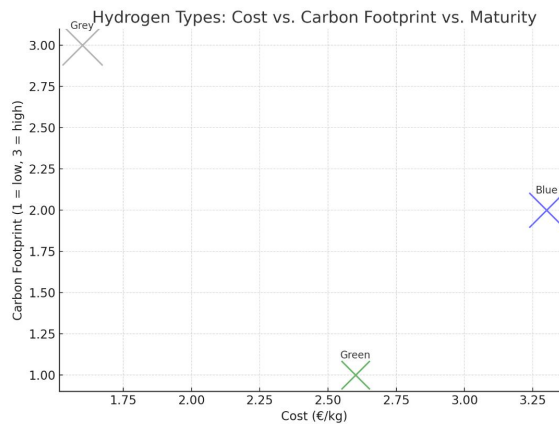


Figure 7: Comparison of Hydrogen Types in Morocco: Cost vs. Carbon Footprint vs. Maturity

The chart in Fig. 7 provides a comparative analysis of three hydrogen types : grey, blue, and green—based on cost (€/kg), carbon footprint, and technological maturity (represented by bubble size).

- **Grey hydrogen** appears on the lower-cost side (€1.6/kg) but carries the highest carbon footprint (score 3/3), as it is produced from natural gas without carbon capture. Despite being mature and cost-effective, it is increasingly incompatible with climate goals.

- **Blue hydrogen**, costing around €3.3/kg, offers a moderate carbon footprint (score 2/3) due to partial CO₂ mitigation through carbon capture and storage (CCS). However, the high cost and infrastructure requirements make it less attractive in the long term.
- **Green hydrogen**, with a cost of about €2.6/kg, is the cleanest option with the lowest carbon footprint (score 1/3). Its production via renewable-powered electrolysis ensures sustainability, though it is still in the scaling-up phase despite growing maturity.

This comparison highlights green hydrogen as a promising pathway due to its low carbon footprint and potential for cost reduction by 2050. While environmental sustainability involves broader factors, such as water use and land impact, green hydrogen stands out for its alignment with long-term decarbonization goals.

3.2 Transport of Hydrogen from Morocco to Hamburg

3.2.1 Transport Modes:

Several options are being considered for exporting Moroccan hydrogen. A subsea pipeline connecting the Moroccan Sahara to Northern Europe has been proposed, but it remains a long-term and costly project (over 6,000 km, estimated at \$2.5–4 million per km) [18].

In the short term, maritime transport is the most viable option due to its lower initial infrastructure costs, greater flexibility, and technological maturity for liquid hydrogen and ammonia shipping. This is supported by cost projections from Aurora Energy Research (2023) and IRES (2023), which show that while pipelines are competitive in the long run, shipping methods are more economical and feasible for initial deployment [16, 13].

The three main maritime transport forms are:

- **Liquid Hydrogen (LH₂)**: Requires liquefaction at –253°C (approximately 10–12 kWh/kg). Although still emerging, this method offers moderate energy losses (around 30–35% of the hydrogen’s energy content) [19].
- **Green Ammonia (NH₃)**: Produced from hydrogen (15 kWh/kg), it carries about 17.6% hydrogen by mass. Transported using existing ammonia infrastructure, it must later be cracked back into hydrogen, leading to high energy losses (80–100 kWh/kg) [13].
- **Liquid Organic Hydrogen Carriers (LOHC)**: Hydrogen is chemically bonded to a liquid (e.g., toluene) and released at the destination. While this method is still less mature, it is considered promising and cost-competitive due to its storage and handling advantages [17].

3.2.2 Energy and Cost Estimates:

According to Aurora Energy Research (2023), by 2030 the cost of importing 1 kg of Moroccan hydrogen to Germany would be:

- $\sim 4.58 \text{ €/kg}$ via LH_2
- $\sim 4.68 \text{ €/kg}$ via LOHC
- $\sim 4.72 \text{ €/kg}$ via NH_3 (including reconversion)

A pipeline, if realised, could reduce this to about $\text{€}3.70/\text{kg}$ [16]. Per 4,000 km, this equates to roughly $\text{€}0.001\text{--}0.002/\text{kg}\cdot\text{km}$ in transport cost [20]. Energy-wise, liquefaction of hydrogen requires approximately $10\text{--}12 \text{ kWh/kg}$ [19], while ammonia (NH_3) synthesis and cracking back to hydrogen can consume around $80\text{--}100 \text{ kWh/kg}$ [13]. Compression for pipelines adds a few kWh/kg depending on the pressure and pipeline design [20].

3.2.3 Infrastructure and Projects:

Morocco is developing large-scale Power-to-X projects (e.g., in Dakhla and Tarfaya), with connections planned to major ports like Tangier, Agadir, and Laâyoune. A pipeline project (e.g., “Nareva Hydrogen”) from Dakhla to Europe is under consideration (estimated at over $\text{\$}20$ billion). The trans-Saharan gas pipeline (Nigeria–Morocco) could also be adapted for hydrogen.

On the German side, the Port of Hamburg is positioning itself as a hydrogen import hub. A green ammonia terminal (Air Products–Mabanaft) is under construction, set to begin operations around 2026. Additional terminals for LH_2 and LOHC are also being studied.

3.2.4 Outlook for 2030–2050:

Hydrogen transport remains costly today, but the expansion of electrolyser gigafactories and marine infrastructure is expected to lower prices. By 2030, imports of green hydrogen from Morocco could become competitive with local German production. By 2050, IRENA projects that 55% of international hydrogen trade will occur via pipelines, $\sim 40\%$ via ammonia, and $\sim 5\%$ via liquid hydrogen. Morocco could emerge as a key exporter provided infrastructure is developed and renewable electricity remains affordable.

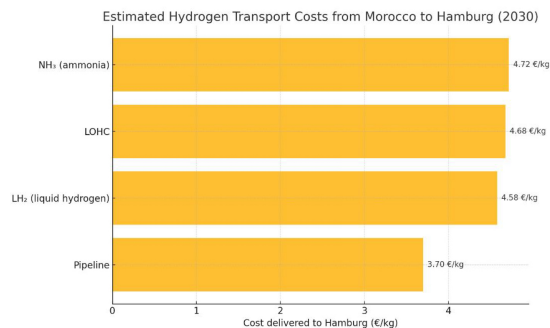


Figure 8: Estimated transport costs per kilogram of hydrogen from Morocco to Hamburg by 2030 for different carriers (NH_3 , LOHC, LH_2 , pipeline).

This bar chart illustrates the estimated hydrogen transport costs from Morocco to Hamburg by 2030, comparing four main delivery methods: ammonia (NH_3), LOHC (Liquid Organic Hydrogen Carrier), liquid hydrogen (LH_2), and pipeline transmission [16].

- **Ammonia (NH_3)** appears as the most expensive option, with a projected delivery cost of €4.72/kg. This method benefits from established global infrastructure but involves complex reconversion steps back to hydrogen.
- **LOHC**, with a transport cost of €4.68/kg, allows hydrogen to be chemically bound to a liquid carrier, offering safer handling but requiring significant energy input for dehydrogenation.
- **Liquid hydrogen (LH_2)**, priced at €4.58/kg, requires cryogenic conditions for transport, making it technically demanding and costly due to boil-off and insulation needs.
- **Pipeline transport** emerges as the most cost-effective solution, with a projected cost of €3.70/kg, assuming the availability of dedicated or retrofitted hydrogen pipelines. However, this option entails substantial upfront infrastructure investment and geopolitical coordination.

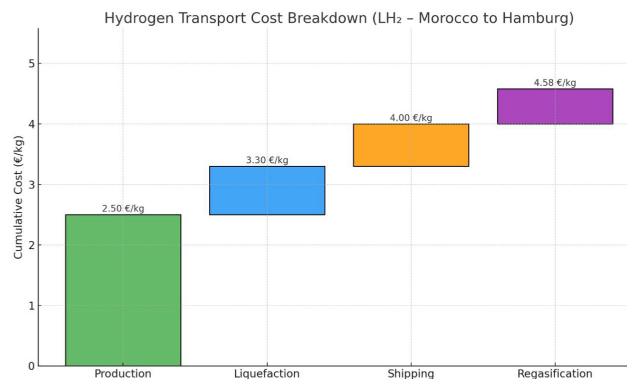


Figure 9: Breakdown of Hydrogen Transport Costs via Liquid Hydrogen (LH_2) from Morocco to Hamburg (2030)

The cumulative cost reaches €4.58/kg, consisting of production (€2.50/kg), liquefaction (€0.80/kg), shipping (€0.70/kg), and regasification (€0.58/kg).

This waterfall chart shows each step in the cost breakdown for transporting 1 kg of liquid hydrogen (LH_2) from Morocco to Hamburg.

Distribution of Renewable Capacity for Green Hydrogen Projects by Region

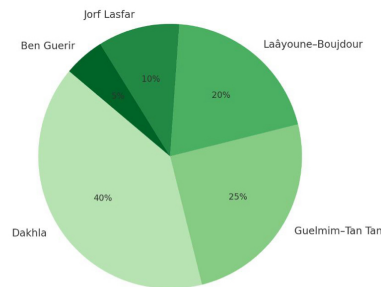


Figure 10: Distribution of Renewable Energy Capacity for Green Hydrogen Projects by Moroccan Region

3.3 Strategic Regions

The pie chart in Fig. 10. illustrates Morocco's strategic prioritization in the allocation of renewable energy capacity for green hydrogen production. Southern regions, particularly Dakhla and Guelmim-Tan Tan, stand out as primary hubs due to their superior solar and wind potential and availability of large, uninhabited land.

This reflects a clear policy trend favoring optimal geoclimatic conditions to ensure high capacity factors and scalability. Meanwhile, regions like Laâyoune-Boujdour, Jorf Lasfar, and Ben Guerir play a complementary role, supporting diversification and risk balancing across the national hydrogen strategy.

3.4 Major Projects in Progress

Example for *Chbika project*:

- Installed capacity: 1,000 MW
- Capacity factor: 40%
- Number of hours in a year: 8,760 hours

Step 1: Theoretical annual generation (without considering the capacity factor):

$$1,000 \text{ MW} \times 8,760 \text{ hours/year} = 8,760,000 \text{ MWh/year}$$

Step 2: Actual annual production (with 40% capacity factor):

$$0.40 \times 8,760,000 = 3,504,000 \text{ MWh/year}$$

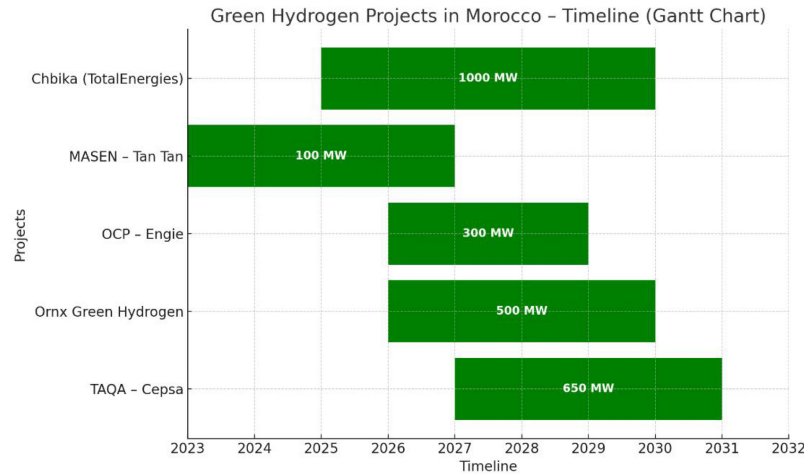


Figure 11: Timeline of Major Green Hydrogen Projects in Morocco with project size (MW)

Table 5: Estimated Capacity Factors for Major Hydrogen Projects in Morocco

Project	Estimated capacity factor
Chbika (Total)	~40–45% (optimized solar–wind mix)
MASEN - Tan Tan	~35–38% (pilot project)
Ornx - Dakhla	~45–48% (ideal wind and solar zone)
OCP-Engie-Jorf	~30% (mostly solar-based)
TAQA-Cepsa-Dakhla	~45% (strong coastal wind and solar mix)

Note: The estimated capacity factors are not explicitly published in project announcements by developers such as TotalEnergies, TAQA, MASEN, or OCP. However, based on the installed capacity and publicly disclosed expected annual output, as well as descriptions of hybrid configurations (solar–wind mix), we can derive approximate values ranging from 30% to 48%. These figures align with assumptions used by [20] and [16] for Moroccan green hydrogen scenarios.

4 Comparative Analysis of Both Scenarios

This analysis compares the two scenarios for hydrogen supply to Hamburg Airport: developing local production capabilities in Northern Germany or importing hydrogen from Morocco. Our analysis reveals a complex trade-off between cost and security. Morocco offers compelling near-term economics with delivered hydrogen at €4.58-4.72/kg compared to €9.59/kg for local production, yet Northern Germany provides superior energy security and environmental credentials. The optimal strategy involves a phased approach, leveraging imports for immediate cost advantages while building local capacity for long-term sustainability.

4.1 Production Landscape

Northern Germany hosts over 21 GW of wind capacity, with 3,384 GWh annually curtailed due to grid constraints – enough to meet over half of Hamburg Airport’s 2040 hydrogen demand. Converting this wasted energy to hydrogen monetizes otherwise lost generation while providing green fuel.

Morocco offers world-class renewable resources with 2,200 kWh/m²/year solar irradiation and 7.5-10 m/s coastal winds, enabling 40%+ capacity factors. Major projects like Chbika (1,000 MW) demonstrate the country’s export ambitions, though the 3,500 km distance to Hamburg poses logistical challenges.

4.2 Technology and Infrastructure

Northern Germany leverages established infrastructure with operational projects including HH-WIN Hamburg (100 MW) and plans for the European Hydrogen Backbone by 2027-2028. Local manufacturers (Siemens Energy, Thyssenkrupp) provide technological advantages, while salt caverns offer storage solutions.

Morocco’s newer sector requires importing most technology, increasing costs. Transport infrastructure demands significant investment for LH₂ (-253°C, 10-12 kWh/kg energy penalty), ammonia synthesis, or LOHC conversion. Hamburg’s Air Products-Mabanaft terminal (2026) represents the first major import infrastructure.

4.3 Economic Proposition

Local production currently costs €9.59/kg, driven by electricity expenses (51.4%) and capital expenditure (26.1%). Morocco achieves €2.60/kg production costs, with delivered prices of €4.58-4.72/kg – a 50% discount. This advantage stems from superior capacity factors and lower costs.

Future projections show costs converging but not reaching parity: local production declining to €3-4/kg by 2050, while Moroccan delivered costs reach €2.5-3/kg, maintaining a persistent advantage.

Table 6: Summary on Comparison of Local vs Import Hydrogen Production

Criteria	Local (Northern Germany)	Import (Morocco)
Production Resources	→ 21 GW installed wind capacity → 3,384 GWh/year curtailed energy → 30% capacity factor	→ 2,200 kWh/m²/year solar → 7.5-10 m/s coastal winds → 40%+ capacity factor
Current H ₂ Cost	→ €9.59/kg	→ €2.60/kg (production) → €4.58-4.72/kg (delivered)
2030 Projected Cost	→ €6-8/kg	→ €3.5-4/kg (delivered)
2050 Projected Cost	→ €3-4/kg	→ €2.5-3/kg (delivered)
Key Infrastructure	→ 100 MW HH-WIN operational → European H₂ Backbone (2027) → Salt cavern storage available	→ Power-to-X projects in development → Requires LH₂/NH₃/LOHC facilities → 3,500 km transport distance
Technology Maturity	→ Established electrolyser industry → Local manufacturers (Siemens, Thyssenkrupp) → Proven grid integration	→ Importing most technology → Developing expertise → Export infrastructure in early stages
Energy Efficiency	→ 95%+ (minimal transport losses)	→ 60-70% (conversion/transport losses)
CO ₂ Emissions	→ <0.5 kg CO ₂ /kg H ₂	→ 0.5-1.0 kg CO ₂ /kg H ₂ (shipping only)
Supply Security	→ High (domestic control)	→ Moderate (geopolitical/maritime risks)
Key Risks	→ Grid congestion → Renewable intermittency → High initial costs	→ Political stability → Supply chain disruption → Currency fluctuation

4.4 Risk and Environmental Considerations

Local production faces technical challenges (renewable intermittency, grid congestion) within a stable regulatory framework, offering high supply security. Imports introduce multiple risks: political stability, maritime transport, and 30-40% efficiency losses in conversion processes.

Environmental impacts strongly favor local production with <0.5 kg CO₂/kg H₂ lifecycle emissions. Imports add 0.5-1.0 kg CO₂/kg from shipping alone, plus indirect emissions from conversion losses requiring 30-40% more renewable electricity. The following table provides a comprehensive summary.

4.5 Strategic Recommendations

4.5.1 Short term (2025-2030)

Importing from Morocco offers compelling economics that can accelerate hydrogen adoption. The €4-5/kg delivered cost enables pilot programmes and initial fleet conversion without prohibitive expenses. This period should focus on securing long-term supply agreements, developing import infrastructure, and gaining operational experience with hydrogen systems.

4.5.2 The medium term (2030-2040)

Should transition toward a hybrid model, targeting 60% imports and 40% local production. This balanced approach provides cost optimisation while building energy security. As local electrolyser costs decline and renewable capacity expands, the economic gap narrows. The European Hydrogen Backbone's completion will enable efficient regional hydrogen trading, potentially accessing low-cost production from across Northern Europe. This period should emphasise developing storage capacity and operational flexibility to manage supply variations.

4.5.3 Long-term (2040+)

Requires continuous reassessment based on evolving conditions. If local renewable capacity expands as projected and electrolyser costs follow predicted learning curves, local production may achieve cost parity with imports. Alternatively, if Morocco successfully develops pipeline infrastructure and maintains political stability, imports could remain economically attractive. The key is maintaining optionality – infrastructure investments should enable multiple supply sources rather than locking in single pathways.

5 Conclusion

The choice between local production and imports for Hamburg Airport's hydrogen supply is a strategic balance that must evolve with market conditions, technological progress, and policy developments. Morocco's current cost advantage offers the opportunity to deliver hydrogen at roughly half the local production cost. This economic reality makes imports attractive for jumpstarting the hydrogen transition and gaining operational experience without prohibitive costs. However, cost alone shouldn't determine strategy. Local production offers intangible benefits – energy security, reduced carbon footprint, regional economic development, and technological leadership – that justify premium pricing during market development. The curtailed renewable energy in Northern Germany represents a unique opportunity to create value from waste, aligning economic and environmental objectives. The recommended phased approach acknowledges these competing priorities. Starting with imports provides immediate access to affordable hydrogen, enabling the airport to develop hydrogen operations and demonstrate a commitment to decarbonisation. Gradually building Local production capacity creates optionality and security while supporting the broader German hydrogen economy.

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